

Comparison of Fourier Series & Fourier Transform

Parameters	Fourier Series	Fourier Transform
Function	$\Rightarrow x(t) = \sum_{n=-\infty}^{\infty} a_n e^{jn\omega_0 t}$	$\Rightarrow x(\omega) = \int_{-\infty}^{\infty} e^{-j\omega t} x(t) dt$
Purpose	▪ Used for Periodic Signals.	▪ Used for General Signals {Periodic & Aperiodic}.
Output	▪ Discrete Spectrum {Sum of Sine and Cosine terms}	▪ Continuous Spectrum
Representation	▪ Sum of harmonics	▪ Frequency Domain Representation

▪ Fourier Series:

$$\Rightarrow x(t) = \sum_{n=-\infty}^{\infty} a_n e^{jn\omega_0 t} = \dots + a_{-2} e^{-j2\omega_0 t} + a_{-1} e^{-j\omega_0 t} + a_0 + a_1 e^{j\omega_0 t} + a_2 e^{j2\omega_0 t} + \dots$$

- Here, a_n describes the magnitude of the Frequency component and $n\omega_0$ describes the frequency of the spectrum.

Fourier Series Co-efficient Calculation

- Fourier Series:

$$\Rightarrow x(t) = \sum_{n=-\infty}^{\infty} a_n e^{jn\omega_0 t}$$

- Here, a_n describes the magnitude of the Frequency component and $n\omega_0$ describes the frequency of the spectrum.

⇒ Multiply $e^{-jk\omega_0 t}$ at both sides

$$\Rightarrow x(t) e^{-jk\omega_0 t} = \sum_{n=-\infty}^{\infty} a_n e^{jn\omega_0 t} e^{-jk\omega_0 t}$$

$$\Rightarrow x(t) e^{-jk\omega_0 t} = \sum_{n=-\infty}^{\infty} a_n e^{j(n-k)\omega_0 t}$$

⇒ Let's Integrate both sides by $\int_0^T$.

$$\Rightarrow \int_0^T x(t) e^{-jk\omega_0 t} dt = \int_0^T \sum_{n=-\infty}^{\infty} a_n e^{j(n-k)\omega_0 t} dt$$

$$\Rightarrow \int_0^T x(t) e^{-jk\omega_0 t} dt = \sum_{n=-\infty}^{\infty} a_n \int_0^T e^{j(n-k)\omega_0 t} dt$$

⇒ From Euler's Formula.

$$\begin{aligned} \Rightarrow \int_0^T e^{j(n-k)\omega_0 t} dt &= \int_0^T \cos(n-k)\omega_0 t dt + j \int_0^T \sin(n-k)\omega_0 t dt \\ &= T \quad \rightarrow \text{For } n = k \\ &= 0 \quad \rightarrow \text{For } n \neq k \end{aligned}$$

⇒ Hence,

$$\Rightarrow a_n T = \int_0^T x(t) e^{-jn\omega_0 t} dt$$

$$\Rightarrow a_n = \frac{1}{T} \int_0^T x(t) e^{-jn\omega_0 t} dt$$

Types of Fourier Series

1. Exponential Fourier Series:

$$\Rightarrow x(t) = \sum_{n=-\infty}^{\infty} a_n e^{jn\omega_0 t} \quad \text{Where, } a_n = \frac{1}{T} \int_0^T x(t) e^{-jn\omega_0 t} dt$$

- Here, a_n describes the magnitude of the Frequency component and $n\omega_0$ describes the frequency of the spectrum.

2. Trigonometric Fourier Series:

$$\Rightarrow x(t) = a_0 + \sum_{n=1}^{\infty} a_n \cos(n\omega_0 t) + \sum_{n=1}^{\infty} b_n \sin(n\omega_0 t)$$

$$\text{Where, } a_0 = \frac{1}{T} \int_0^T x(t) dt, a_n = \frac{2}{T} \int_0^T x(t) \cos(n\omega_0 t) dt, b_n = \frac{2}{T} \int_0^T x(t) \sin(n\omega_0 t) dt$$

- Here, a_0 describes the DC magnitude of the signal.

3. Polar Fourier Series:

- Polar Fourier Series is derived from Trigonometric Fourier Series by combining the sine and cosine terms of same frequency.

$$\Rightarrow x(t) = C_0 + \sum_{n=1}^{\infty} C_n \cos(n\omega_0 t + \phi_n)$$

$$\text{Where, } C_0 = a_0, C_n = \sqrt{a_n^2 + b_n^2}, \phi_n = -\tan^{-1} \left(\frac{b_n}{a_n} \right)$$

Properties of Fourier Series

□ Linearity Property: If $x_1(t)$ and $x_2(t)$ have Fourier Series F_{n1} and F_{n2} , then

$$\Rightarrow a_1 x_1(t) + a_2 x_2(t) \xleftrightarrow{\text{F. S.}} a_1 F_{n1} + a_2 F_{n2}$$

□ Time Shifting Property: If $x(t)$ has Fourier Series F_n , then

$$\Rightarrow x(t - t_0) \xleftrightarrow{\text{F. S.}} e^{-jn\omega_0 t_0} F_n$$

$$\Rightarrow x(t) = \sum_{n=-\infty}^{\infty} a_n e^{jn\omega_0 t}$$

□ Frequency Shifting Property: If $x(t)$ has Fourier Series F_n , then

$$\Rightarrow e^{jn_0 \omega_0 t} x(t) \xleftrightarrow{\text{F. S.}} F_{(n-n_0)} \quad \Rightarrow e^{-jn_0 \omega_0 t} x(t) \xleftrightarrow{\text{F. S.}} F_{(n+n_0)}$$

□ Time Scaling Property: If $x(t)$ has Fourier Series F_n , then

$$\Rightarrow x(at) \xleftrightarrow{\text{F. S.}} F_n = \sum_{n=-\infty}^{\infty} a_n e^{jn\omega_0 t/a}$$

□ Multiplication & Convolution Property:

$$\Rightarrow x_1(t) * x_2(t) \xleftrightarrow{\text{F. S.}} F_n = \sum_{n=-\infty}^{\infty} a_{n1} a_{n2} e^{jn\omega_0 t}$$

❑ Differentiation Property: If $x(t)$ has Fourier Series F_n , then

$$\Rightarrow \frac{dx(t)}{dt} \xleftrightarrow{\text{F.S.}} F'_n = \sum_{n=-\infty}^{\infty} jn\omega_0 a_n e^{jn\omega_0 t}$$

$$\Rightarrow x(t) = \sum_{n=-\infty}^{\infty} a_n e^{jn\omega_0 t}$$

❑ Integration Property: If $x(t)$ has Fourier Series F_n , then

$$\Rightarrow \int x(t) dt \xleftrightarrow{\text{F.S.}} F'_n = \sum_{n=-\infty}^{\infty} \frac{a_n}{jn\omega_0} e^{jn\omega_0 t}$$

❑ Symmetry Property {Even Function}: If $x(t)$ has Fourier Series F_n , then

$$\Rightarrow a_n^* = a_n \text{ \{Here Fourier Coefficient will be real\} \{Only Cosine Terms, } b_n = 0 \}$$

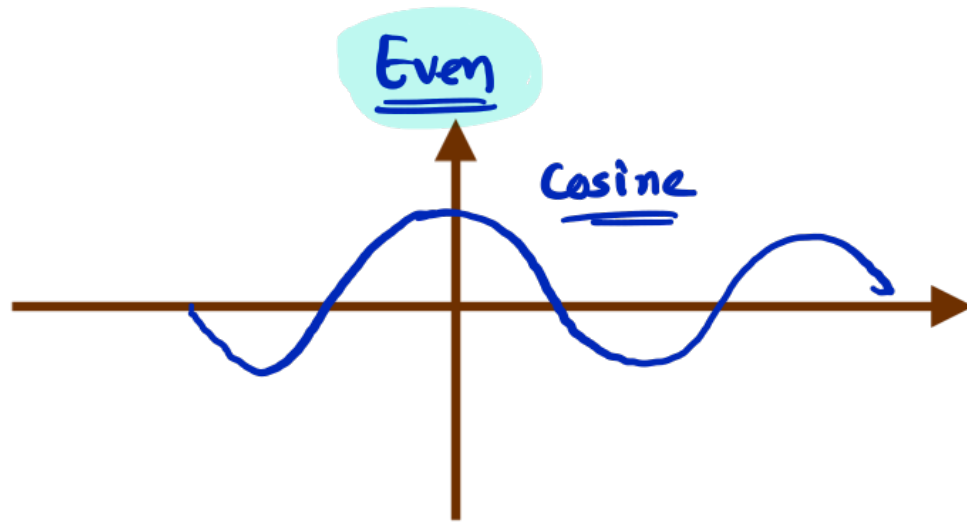
❑ Symmetry Property {Odd Function}: If $x(t)$ has Fourier Series F_n , then

$$\Rightarrow a_n^* = -a_n \text{ \{Here Fourier Coefficient will be imaginary\} \{Only Sine Terms, } a_0 = a_n = 0 \}$$

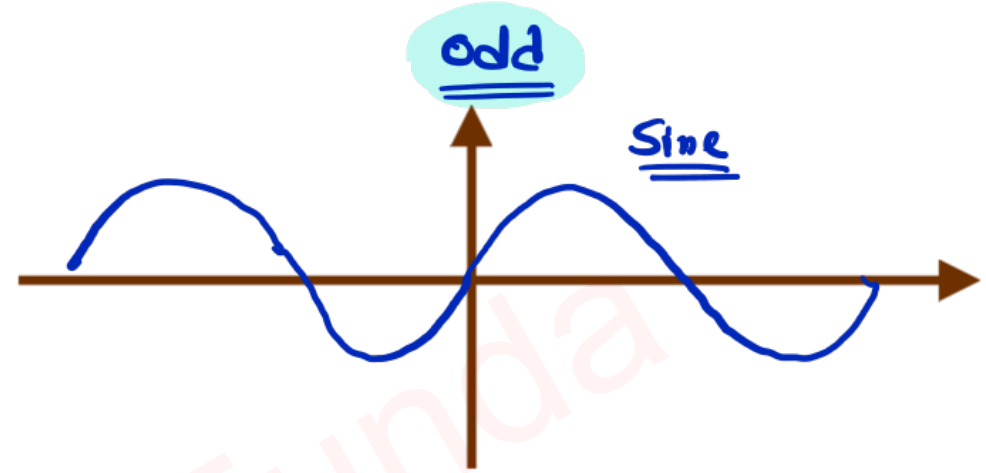
$$\Rightarrow a_n = \frac{1}{T} \int_0^T x(t) e^{-jn\omega_0 t} dt$$

$$\leadsto a = \underbrace{b}_{\text{Real}} + j \underbrace{c}_{\text{Imag}} \Rightarrow a^* = \underbrace{b}_{\text{Real}} - j \underbrace{c}_{\text{Imag.}}$$

Conjugate of a



- Even functions has only Cosine components.



- Odd functions has only Sine components.

Relation between Trigonometric Fourier Series & Exponential Fourier Series

⇒ Trigonometric Fourier Series

$$\Rightarrow x(t) = a_0 + \sum_{n=1}^{\infty} a_n \cos(n\omega_0 t) + \sum_{n=1}^{\infty} b_n \sin(n\omega_0 t) \quad \text{--- (1)}$$

⇒ Exponential Fourier Series

$$\begin{aligned} \Rightarrow x(t) &= \sum_{n=-\infty}^{\infty} C_n e^{jn\omega_0 t} \\ &= C_0 + C_1 e^{j\omega_0 t} + C_2 e^{j2\omega_0 t} + \dots \\ &\quad + C_{-1} e^{-j\omega_0 t} + C_{-2} e^{-j2\omega_0 t} + \dots \end{aligned}$$

$$[e^{jn\omega_0 t} = \cos(n\omega_0 t) + j \sin(n\omega_0 t) , e^{-jn\omega_0 t} = \cos(n\omega_0 t) - j \sin(n\omega_0 t)]$$

$$\Rightarrow x(t) = C_0 + C_1 [\cos(\omega_0 t) + j \sin(\omega_0 t)] + C_2 [\cos(2\omega_0 t) + j \sin(2\omega_0 t)] + \dots$$

$$+ c_{-1} [\cos(\omega t) - j \sin(\omega t)] + c_{-2} [\cos(2\omega t) - j \sin(2\omega t)] + \dots$$

$$= c_0 + \sum_{n=1}^{\infty} (c_n + c_{-n}) \cos(n\omega t) + j \sum_{n=1}^{\infty} (c_n - c_{-n}) \sin(n\omega t) \quad \text{--- ①}$$

$$\leadsto c_0 = a_0$$

$$a_n = c_n + c_{-n}$$

$$b_n = j(c_n - c_{-n})$$

$$ja_n = jc_n + jc_{-n}$$

$$b_n = jc_n - jc_{-n}$$

$$\Rightarrow ja_n + b_n = 2jc_n$$

$$\Rightarrow c_n = \frac{1}{2} (a_n - jb_n)$$

$$ja_n = jc_n + jc_{-n}$$

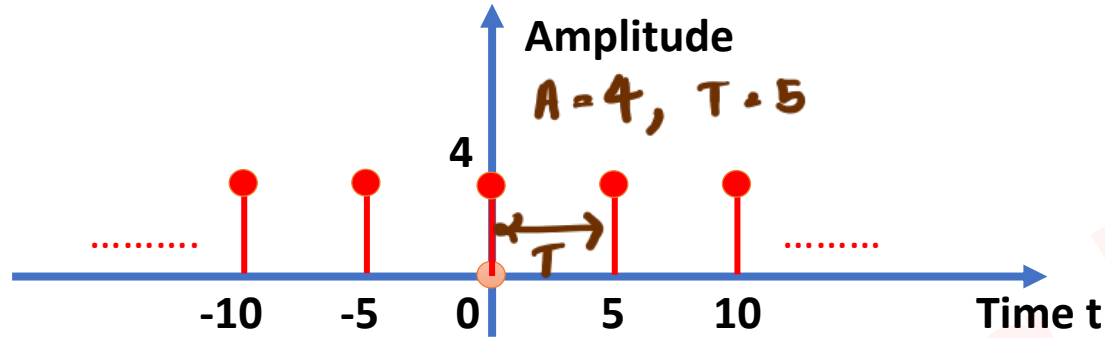
$$-b_n = jc_n - jc_{-n}$$

$$\Rightarrow ja_n - b_n = 2jc_{-n}$$

$$\Rightarrow c_{-n} = \frac{1}{2} (a_n + jb_n)$$

Examples of Fourier Series Coefficients

Example 1. Find the Fourier Series Coefficient of the given signal.



- ✓ a. $4/5$ b. $5/4$ c. $\cos \pi/4$ d. $\sin \pi/4$

$$\begin{aligned} \Rightarrow C_n &= \frac{1}{T} \int_{-T/2}^{T/2} x(t) e^{-j\omega_n t} dt \\ &= \frac{1}{5} \int_{-T/2}^{T/2} (A \delta(t)) e^{-j\omega_n t} dt \\ &= \frac{1}{5} [4 \times 1 \times e^0] \\ &= 4/5 \end{aligned}$$

Example 2. If $x(n)$ is real and even signal then Fourier series coefficient will be

- a. Imaginary and Odd
- b. Imaginary and Even
- c. Real and Odd
- ✓ d. Real and Even

- Signal

Odd & Real

Even & Real

Even & Imag

Odd & Imag.

- Coefficient

- Odd & Imag.

- Even & Real.

- Even & Imag.

- Odd & Real.

Example 3. Find Fourier Series Coefficient for $\cos^2 t$.

a. $-\frac{1}{4}\delta(k-2) + \frac{1}{2}\delta(k) - \frac{1}{4}\delta(k+2)$

b. $+\frac{1}{4}\delta(k-2) - \frac{1}{2}\delta(k) + \frac{1}{4}\delta(k+2)$

✓ c. $+\frac{1}{4}\delta(k-2) + \frac{1}{2}\delta(k) + \frac{1}{4}\delta(k+2)$

d. None

$$\begin{aligned}\leadsto \cos^2 t &= \frac{1 + \cos 2t}{2} \\ &= \frac{1}{2} + \frac{1}{2} \left(\frac{e^{j2t} + e^{-j2t}}{2} \right) \\ &= \frac{1}{2} + \frac{1}{4} e^{j2t} + \frac{1}{4} e^{-j2t}\end{aligned}$$

$\begin{matrix} \uparrow & \uparrow & \uparrow \\ n=0 & n=2 & n=-2 \\ C_0 = 1/2 & C_2 = 1/4 & C_{-2} = 1/4 \end{matrix}$

$$\leadsto C_n = \frac{1}{2} \delta(k) + \frac{1}{4} \delta(k-2) + \frac{1}{4} \delta(k+2)$$

Fourier Series Coefficients using Symmetry

1. Even Symmetry of Function/Signal:

$$\Rightarrow x(t) = x(-t)$$

- Only a_n {DC and Cosine Terms} Exists. $b_n = 0$ {Sine Terms are Zero}

2. Odd Symmetry of Function/Signal:

$$\Rightarrow x(t) = -x(-t)$$

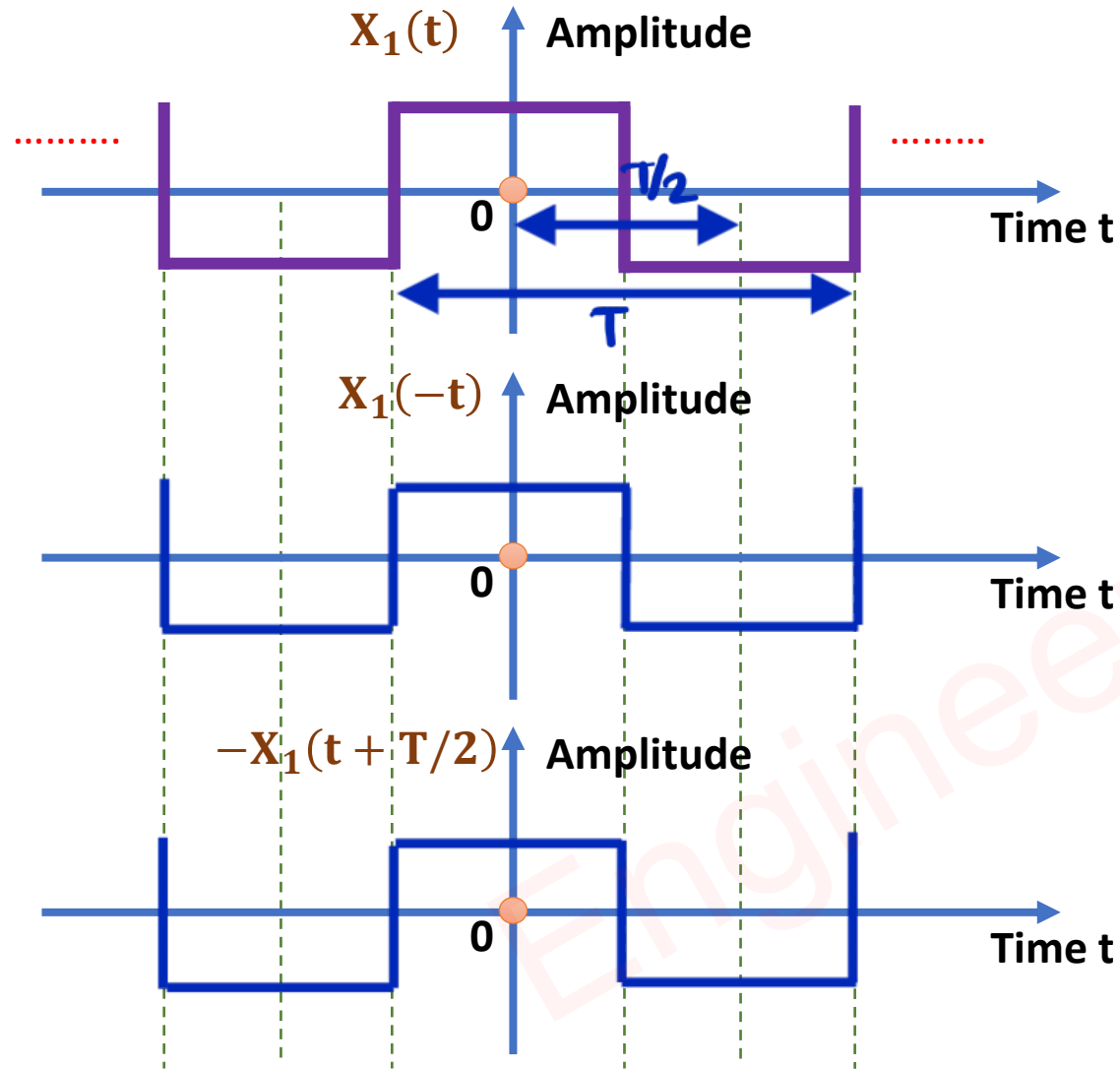
- Only b_n {Sine Terms} Exists. $a_n = 0$ {Cosine Terms are Zero}

3. Half-Wave Symmetry of Function/Signal:

$$\Rightarrow x(t) = -x(t + T/2)$$

- Only Odd Harmonics of a_n and b_n is available.

Example – 1: $X_1(t)$



$$\leadsto x_1(t) = x_1(-t)$$

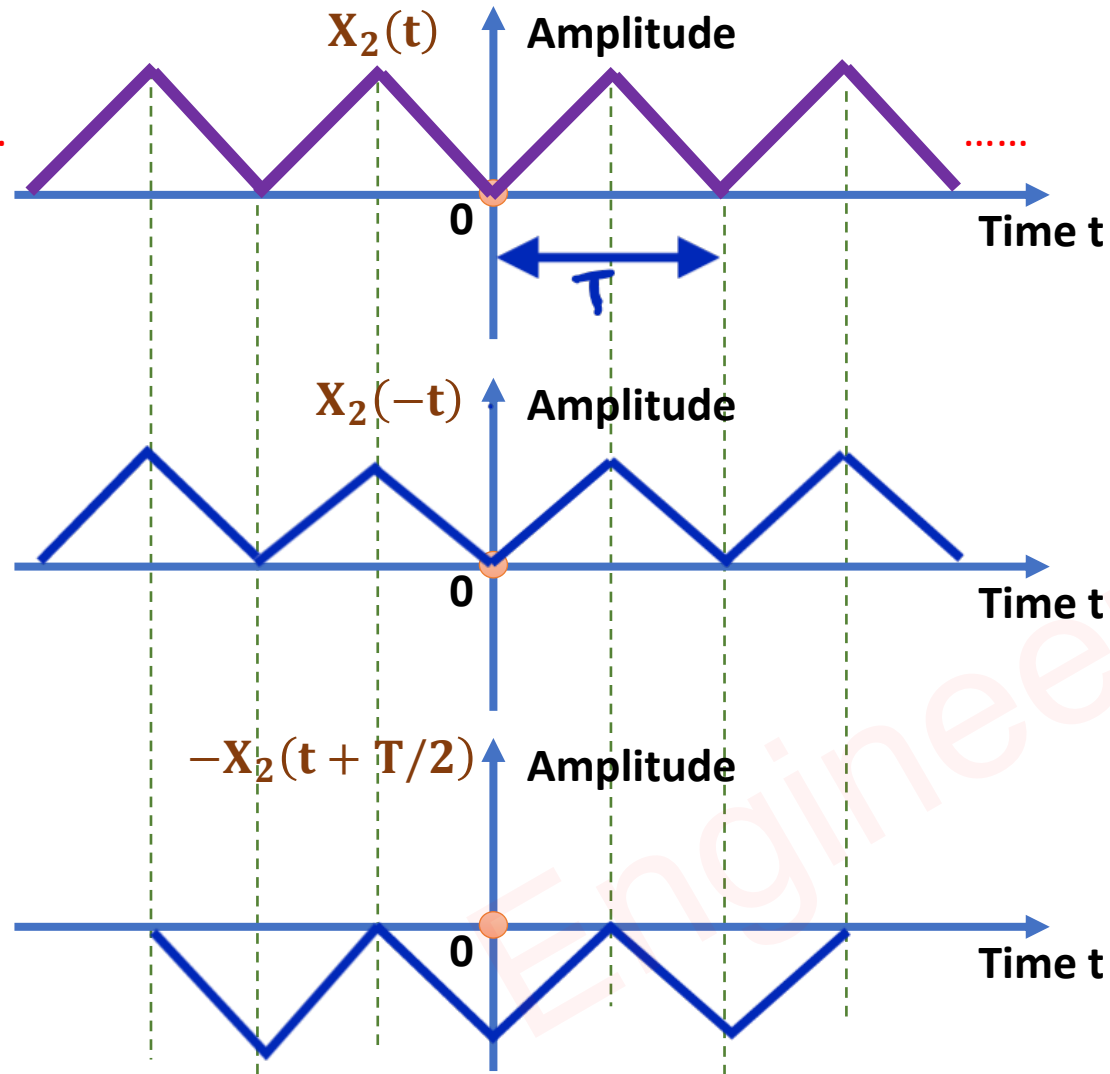
- Even Symmetry
- Only Cosine terms.

$$\leadsto x_1(t) = -x_1(t + T/2)$$

- Half-wave symmetry.
- Odd Harmonics.

$$- \cos \omega_0 t, \cos 3\omega_0 t, \cos 5\omega_0 t, \dots$$

Example – 2: $X_2(t)$



$$\leadsto x_2(t) = x_2(t+T)$$

– Even Symmetry.

– Only Cosine terms.

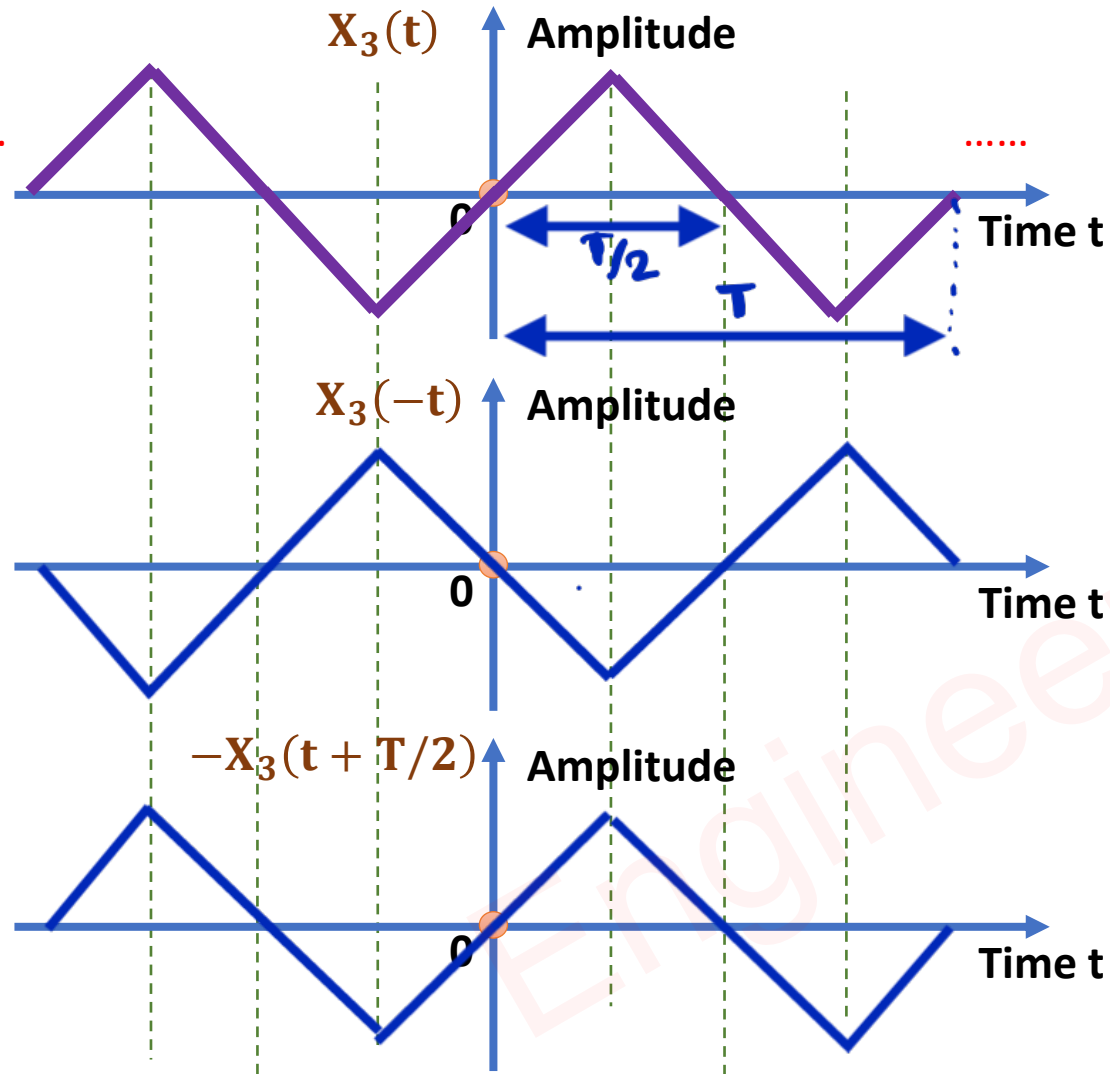
$$\leadsto x_2(t) \neq -x_2(t+T/2)$$

– It does not follow Half-wave Symmetry

– all terms are available.

– $\cos \omega t$, $\cos 2\omega t$, $\cos 3\omega t$, ...

Example – 3: $X_3(t)$



$$\leadsto x_3(t) = -x_3(-t)$$

- odd symmetry

- Only sine terms.

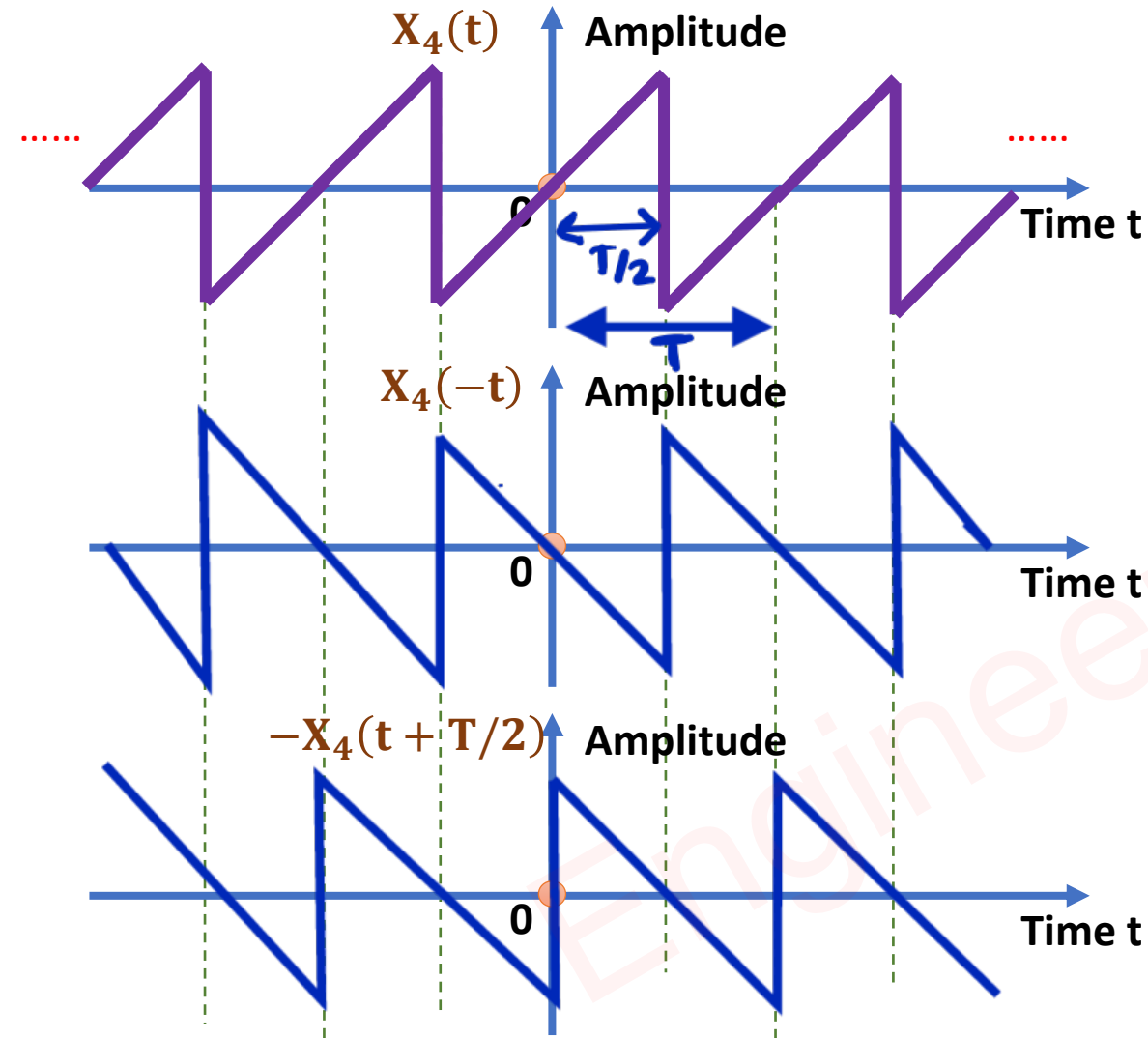
$$\leadsto x_3(t) = -x_3(t + T/2)$$

- Half-wave symmetry.

- Only odd harmonics.

- $\sin \omega t$, $\sin 3\omega t$, $\sin 5\omega t$, ...

Example – 4: $X_4(t)$



$$\Rightarrow x_4(t) = -x_4(-t)$$

- Odd Symmetry.

- Only sine terms.

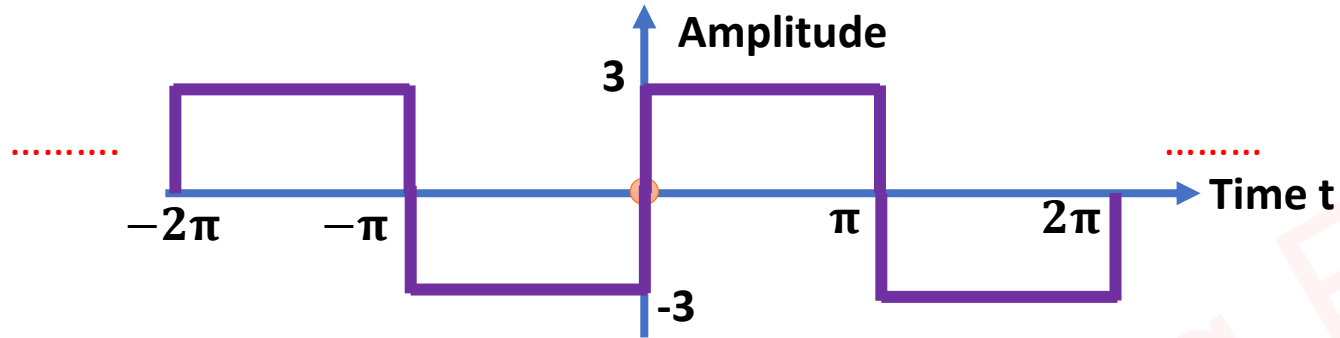
$$\Rightarrow x_4(t) \neq -x_4(t + T/2)$$

- all Harmonics.

- $\sin \omega_0 t, \sin 2\omega_0 t, \sin 3\omega_0 t, \dots$

Examples of Fourier Series

Example 1. Find the Fourier Series of the given signal.



$$\leadsto x(t) = -x(-t)$$

- Odd signal

- Only Sine Harmonics

$$\leadsto x(t) = -x(t + T/2)$$

- Half-wave symmetry

- Only odd Harmonics

a. $\frac{3}{2} + \frac{2}{\pi} \left[\cos t + \frac{1}{3} \cos 3t + \frac{1}{5} \cos 5t + \dots \right]$

b. $\frac{3}{2} + \frac{12}{\pi} \left[\sin t + \frac{1}{2} \sin 2t + \frac{1}{2} \sin 3t + \dots \right]$

c. $\frac{12}{\pi} \left[\cos t + \frac{1}{3} \cos 3t + \frac{1}{5} \cos 5t + \dots \right]$

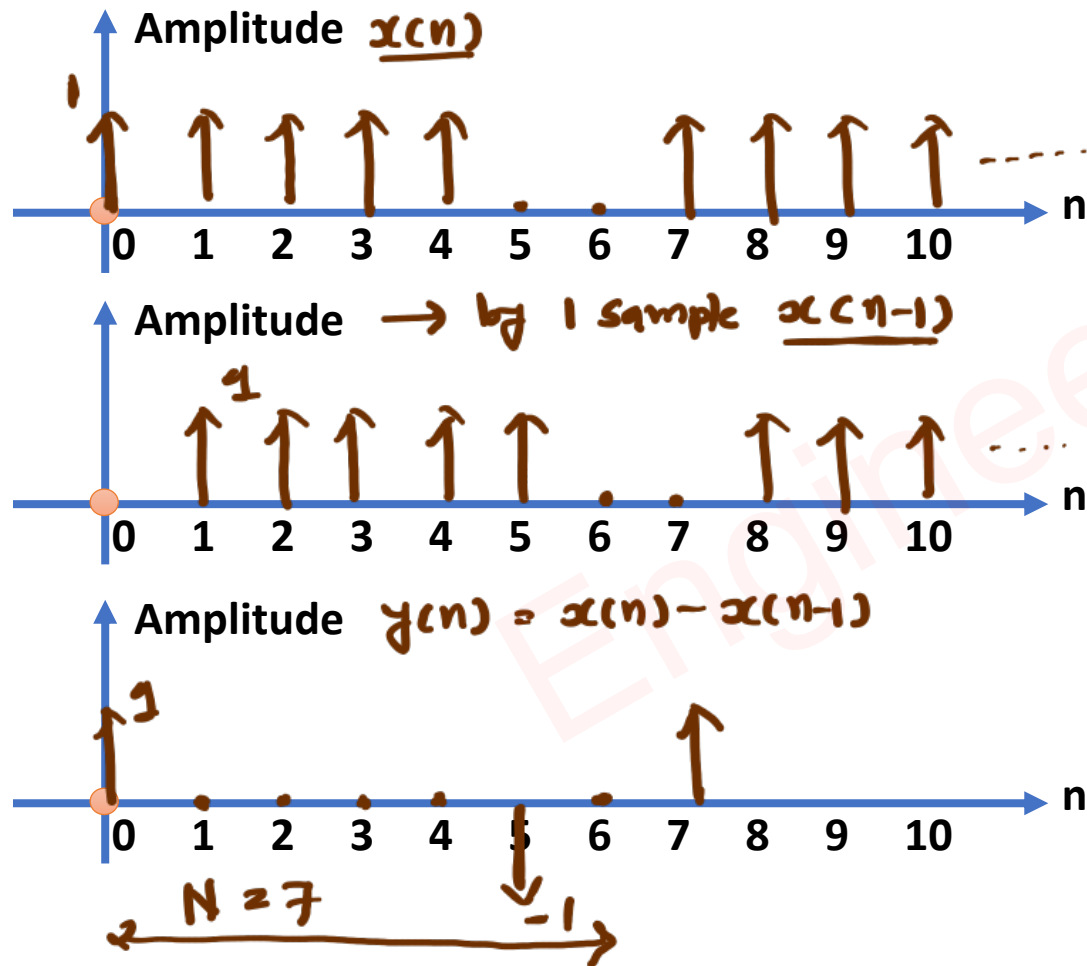
✓ d. $\frac{12}{\pi} \left[\sin t + \frac{1}{3} \sin 3t + \frac{1}{5} \sin 5t + \dots \right]$

Example 2. Given Discrete Time Periodic Signal

$$x(n) = \begin{cases} 1 & 0 \leq n \leq 4 \\ 0 & 5 \leq n \leq 6 \end{cases}$$

a. Fundamental Period of $y(n) = x(n) - x(n-1)$

b. Find Fourier Series Coefficient of $x(n)$.



$$\begin{aligned} \sim [C_k] &= \frac{1}{N} \sum_{n=0}^{N-1} x(n) e^{-j\omega_0 kn} \\ &= \frac{1}{N} \sum_{n=0}^{N-1} x(n) e^{-j\left(\frac{2\pi}{N}\right)kn} \\ &= \frac{1}{7} \sum_{n=0}^6 x(n) e^{-j\left(\frac{2\pi}{7}\right)kn} \\ &= \frac{1}{7} \sum_{n=0}^4 e^{-j\left(\frac{2\pi}{7}\right)kn} \\ &= \left[\sum_{n=0}^m r^n = \frac{1-r^{m+1}}{1-r} \right] \\ \sim [C_k] &= \frac{1}{7} \left[\frac{1-e^{-j\left(\frac{10\pi}{7}\right)k}}{1-e^{-j\left(\frac{2\pi}{7}\right)k}} \right] \end{aligned}$$

Example 3. Exponential Fourier Series Coefficient is given by $C_2 = 1/2$ and Phase Angle is 45° . Then Find Trigonometric Fourier Series Coefficient a_2 and b_2 , Also identify Average Power of the signal.

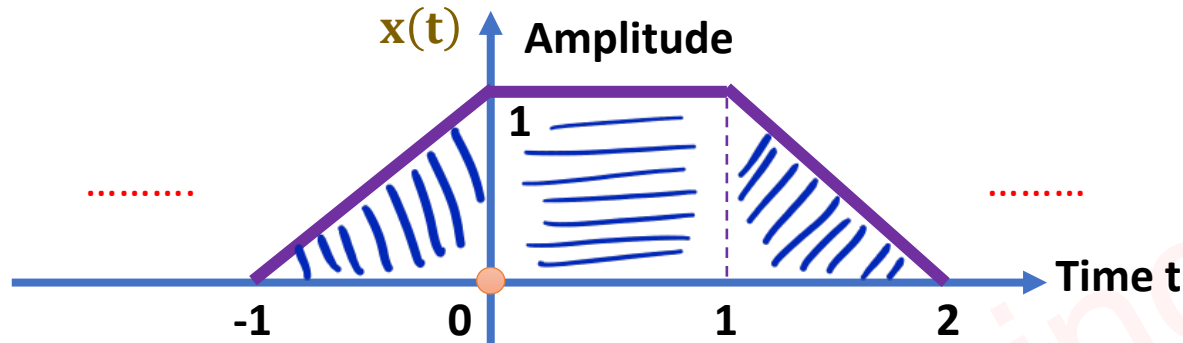
$$\begin{aligned}\leadsto a_2 &= 2 |C_2| \cos \theta_2 \\ &= 2 \left| \frac{1}{2} \right| \cos 45^\circ \\ &= 1/\sqrt{2}\end{aligned}$$

$$\begin{aligned}\leadsto b_2 &= 2 |C_2| \sin \theta_2 \\ &= 2 \left| \frac{1}{2} \right| \sin 45^\circ \\ &= 1/\sqrt{2}\end{aligned}$$

$$\begin{aligned}\leadsto P_{avg} &= C_0^2 + 2(C_1^2 + C_2^2 + C_3^2 + \dots + C_n^2) \\ &= 2\left(\frac{1}{2}\right)^2 \\ &= 1/2\end{aligned}$$

Examples of DC Component of Fourier Series

Example 1. Consider we have periodic Signal $x(t)$. Find the DC Component of Fourier Series of the given signal.



↪ DC Component

$$\Rightarrow a_0 = \frac{1}{T} \int_{-1}^2 x(t) dt$$

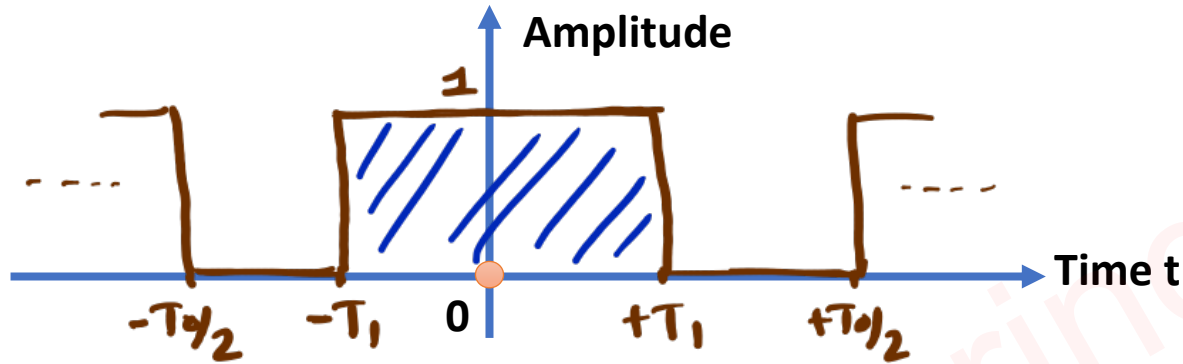
$$= \frac{\text{Area of } x(t) \text{ over } T}{\text{Time period } T}$$

$$= \frac{\frac{1}{2} \times 1 \times 1 + 1 \times 1 + \frac{1}{2} \times 1 \times 1}{3} = \frac{2}{3}$$

Example 2. A Periodic Signal $x(t)$ of period T_0 is given by:

$$x(t) = \begin{cases} 1 & |t| < T_1 \\ 0 & T_1 < |t| < T_0/2 \end{cases}$$

Find DC Component of $x(t)$.



- DC Component.

$$\Rightarrow a_0 = \frac{1}{T} \int_{-T_0/2}^{T_0/2} x(t) dt$$

$$= \frac{\text{Area of } x(t) \text{ over } T}{\text{Time period } T}$$

$$= \frac{2T_1}{T_0}$$

Fundamental Component & Harmonics of Fourier Series

Example 1. Find the amplitude of Fundamental Component of the given signal.

$$\Rightarrow \mathbf{x(t)} = \underbrace{3 \sin(6t + 30^\circ)}_{\omega_1 = 6 \text{ rad/sec}} + \underbrace{4 \cos(12t - 60^\circ)}_{\omega_2 = 12 \text{ rad/sec}}$$

$$\Rightarrow x(t) = \text{DC Component} + 1^{\text{st}} \text{ Harmonic} + 2^{\text{nd}} \text{ Harmonic} + 3^{\text{rd}} \text{ Harmonic} + \dots$$

$\underline{\underline{\text{freq} = 0}}$

Fundamental
Component
 (ω_0)

$(2\omega_0)$

$(3\omega_0)$

- $\text{GCD}[6, 12] = 6 = \omega_0$

- Fundamental Component Amplitude = 3

Example 2. Find the amplitude of Fundamental Component and 2nd, 3rd & 4th Harmonics of the given signal.

$$\Rightarrow x(t) = \underbrace{2 \sin(3t + 15^\circ)}_{\omega_1 = 3 \text{ rad/sec}} + \underbrace{6 \sin(9t + 30^\circ)}_{\omega_2 = 9 \text{ rad/sec}} + \underbrace{8 \cos(12t + 45^\circ)}_{\omega_3 = 12 \text{ rad/sec}}$$

$$\Rightarrow \omega_0 = \text{GCD} [3, 9, 12] = 3 \text{ rad/sec}$$

$$\text{— 1st Harmonic} = \omega_0 = 3 \text{ rad/sec} \rightarrow A_1 = 2$$

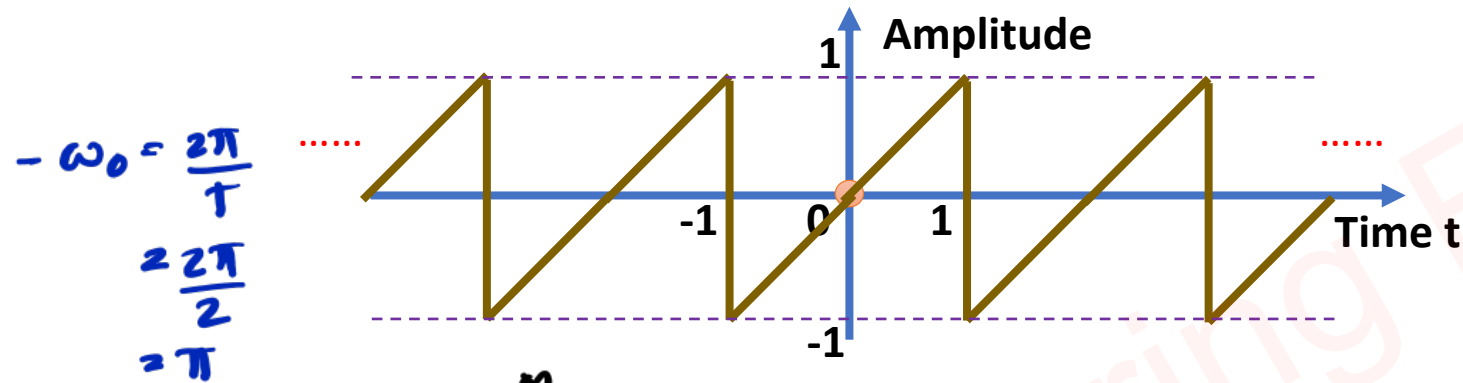
$$\text{2nd " } = 2\omega_0 = 6 \text{ rad/sec} \rightarrow A_2 = 0$$

$$\text{3rd " } = 3\omega_0 = 9 \text{ rad/sec} \rightarrow A_3 = 6$$

$$\text{4th " } = 4\omega_0 = 12 \text{ rad/sec} \rightarrow A_4 = 8$$

Examples of Trigonometric Fourier Series

Example 1. Consider we have periodic Signal $x(t)$. Find the Trigonometric Fourier Series of the given signal.



$$\begin{aligned} \Rightarrow x(t) &= a_0 + \sum_{n=1}^{\infty} a_n \cos(n\omega_0 t) + \sum_{n=1}^{\infty} b_n \sin(n\omega_0 t) \\ &= \sum_{n=1}^{\infty} b_n \sin(n\omega_0 t) \end{aligned}$$

$$\begin{aligned} \Rightarrow b_n &= \frac{2}{T} \int_{-T/2}^{T/2} x(t) \sin(n\omega_0 t) dt \\ &= \frac{2}{2} \int_{-1}^{1} t \sin(n\pi t) dt \end{aligned}$$

$$\begin{aligned} \Rightarrow a_0 &= \frac{1}{T} \int_T x(t) dt \\ &= \frac{\text{Area of } x(t) \text{ w.r.t time}}{\text{Time } T} \\ &= 0 \end{aligned}$$

$$\begin{aligned} \Rightarrow x(t) &= -x(-t) \\ &\text{- Odd Signal} \\ \Rightarrow a_n &= 0 \end{aligned}$$

$$\left[\int u v dx = u \int v dx - \int \left(\frac{du}{dx} \int v dx \right) dx \right]$$

$$\Rightarrow b_n = \left[t \int \sin(n\pi t) dt - \int \int \sin(n\pi t) dt dt \right]_{-1}^1$$

$$= \left[t \left(-\frac{\cos(n\pi t)}{n\pi} \right) - \int \left(-\frac{\cos(n\pi t)}{n\pi} \right) dt \right]_{-1}^1$$

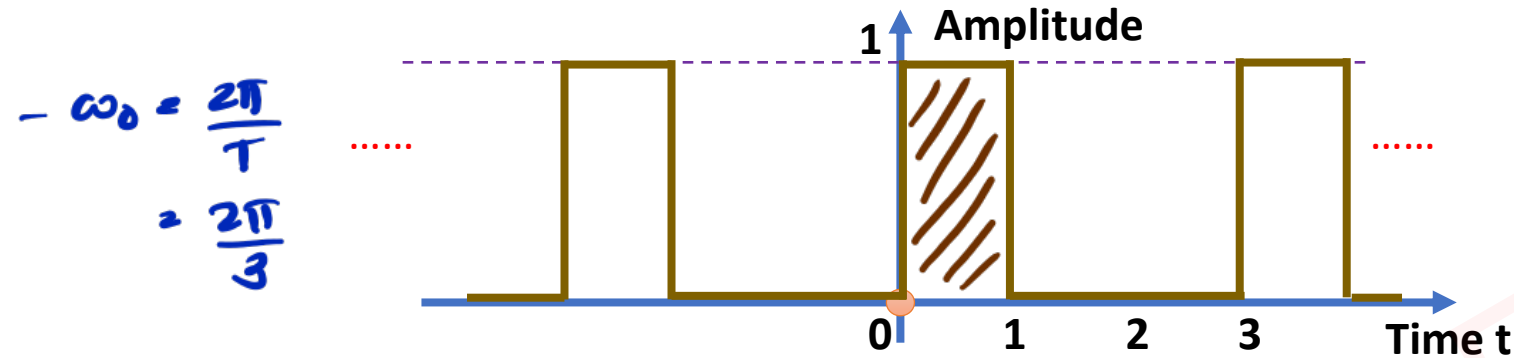
$$= \left[t \left(-\frac{\cos(n\pi t)}{n\pi} \right) + \frac{\sin(n\pi t)}{n^2 \pi^2} \right]_{-1}^1$$

$$= \left[-\frac{1}{n\pi} \cos(n\pi) - \frac{1}{n\pi} \cos(n\pi) \right]$$

$$= -\frac{2}{n\pi} \cos(n\pi)$$

$$\Rightarrow x(t) = \sum_{n=1}^{\infty} \left(-\frac{2}{n\pi} \right) \cos(n\pi) \sin(n\pi t)$$

Example 2. Consider we have periodic Signal $x(t)$. Find the Trigonometric Fourier Series of the given signal.



$$x(t) = a_0 + \sum_{n=1}^{\infty} a_n \cos(n\omega_0 t) + \sum_{n=1}^{\infty} b_n \sin(n\omega_0 t)$$

$$\begin{aligned}
 -a_0 &= \frac{1}{T} \int_T x(t) dt \\
 &= \frac{\text{Area of } x(t) \text{ for } T}{\text{Time } T} \\
 &= \frac{1}{3}
 \end{aligned}$$

$$\begin{aligned}
 a_n &= \frac{2}{T} \int_T x(t) \cos(n\omega_0 t) dt \\
 &= \frac{2}{3} \int_0^1 \cos\left(n\left(\frac{2\pi}{3}\right)t\right) dt \\
 &= \frac{2}{3} \left[\frac{\sin\left(n\left(\frac{2\pi}{3}\right)t\right)}{n\left(\frac{2\pi}{3}\right)} \right]_0^1 \\
 &= \frac{1}{n\pi} \times \sin\left(n\left(\frac{2\pi}{3}\right)\right)
 \end{aligned}$$

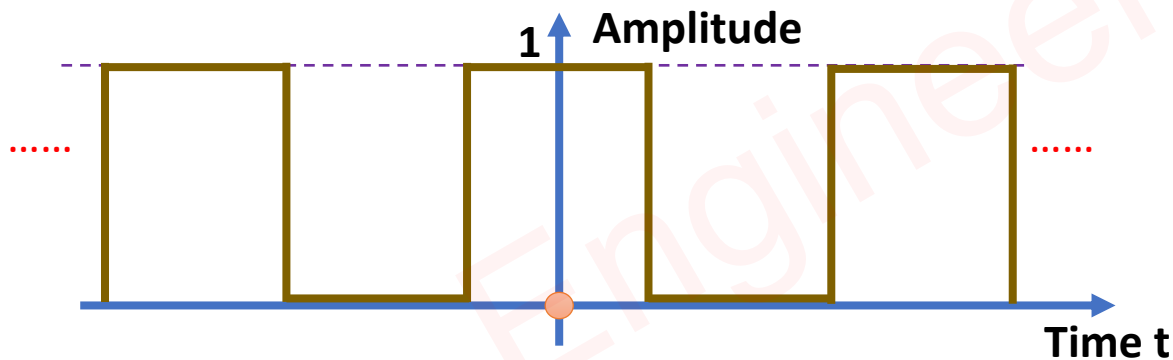
$$\begin{aligned}
 \leadsto b_n &= \frac{2}{T} \int_T x(t) \sin(n\omega_0 t) dt \\
 &= \frac{2}{3} \int_0^1 \sin\left(n\left(\frac{2\pi}{3}\right)t\right) dt \\
 &= \frac{2}{3} \left[-\frac{\cos\left(n\left(\frac{2\pi}{3}\right)t\right)}{n\frac{2\pi}{3}} \right]_0^1 \\
 &= \frac{1}{n\pi} \left[1 - \cos\left(\frac{2\pi n}{3}\right) \right]
 \end{aligned}$$

$$\leadsto x(t) = \frac{1}{3} + \sum_{n=1}^{\infty} \frac{1}{n\pi} \sin\left(n\left(\frac{2\pi}{3}\right)\right) \cos\left(n\left(\frac{2\pi}{3}\right)t\right) + \sum_{n=1}^{\infty} \frac{1}{n\pi} \left[1 - \cos\left(\frac{2\pi n}{3}\right) \right] \times \sin\left(n\left(\frac{2\pi}{3}\right)t\right)$$

Examples of Trigonometric Fourier Series

Example 1. The Fourier expansion $f(t) = a_0 + \sum a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t)$ of the Trigonometric Fourier Series of the periodic signal shown below will contain the following non zero terms:

- a. a_0 and $b_n, n = 1, 3, 5, \dots, \infty$
- b. a_0 and $a_n, n = 1, 2, 3, \dots, \infty$
- c. a_0, a_n and $b_n, n = 1, 2, 3, \dots, \infty$
- d. a_0 and $b_n, n = 1, 3, 5, \dots, \infty$



$\Rightarrow$ DC Component a_0

$$\Rightarrow a_0 = \frac{1}{T} \int_T x(t) dt$$

$$= \frac{\text{Area of } x(t) \text{ over } T}{\text{Time } T}$$

$\neq 0$

$$\Rightarrow x(t) = x(-t)$$

- Even signal. $\Rightarrow b_n = 0, \underline{a_n \neq 0}$

$\Rightarrow$ Half-wave Symmetry

$$\Rightarrow x(t) \neq -x(t + T/2) \leftarrow \text{No H-}\omega\text{-Symmetry}$$

Example 2. $x(t)$ is a real valued function of a real variable with period T . Its Trigonometric Fourier Series Expansion contains no terms of frequency $\omega = 2\pi(2k)/T$; $k = 1, 2, \dots$. Also, No sine terms are present. Then $x(t)$ satisfies the equation:

a. $x(t) = -x(t - T)$ ✗

b. $x(t) = x(T - t) = -x(-t)$ ✗

c. $x(t) = x(T - t) = -x(t - T/2)$ ✓

d. $x(t) = x(t - T) = x(t - T/2)$ ✗

⇒ No Sine terms

- $b_n = 0$

- Even signal.

⇒ $x(t) = x(-t)$

⇒ Freq. not available.

$$\Rightarrow \omega = \frac{2\pi(2k)}{T}, \quad k = 1, 2, 3, \dots$$

$$= \omega_0(2k), \quad k = 1, 2, 3, \dots$$

$$= 2\omega_0, 4\omega_0, 6\omega_0, \dots$$

$$\Rightarrow x(t) = x(-t) = x(-t + T) = -x(t + T/2)$$

⇒ Freq. available.

$$\Rightarrow \omega = \omega_0, 3\omega_0, 5\omega_0, \dots$$

$$\Rightarrow x(t) = -x(t + T/2)$$

Example 3. Prove that $x(t)$ is periodic signal.

$$x(t) = a_0 + \sum_{n=1}^{\infty} [a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t)]$$

⇒ For periodic signal

$$\Rightarrow x(t) = x(t+T)$$

$$\Rightarrow x(t+T) = a_0 + \sum_{n=1}^{\infty} (a_n \cos(n\omega_0(t+T)) + b_n \sin(n\omega_0(t+T)))$$

$$= a_0 + \sum_{n=1}^{\infty} (a_n \cos(n(\omega_0 t + \omega_0 T)) + b_n \sin(n(\omega_0 t + \omega_0 T)))$$

$$\Rightarrow \omega_0 T = \frac{2\pi}{T} \times T = 2\pi$$

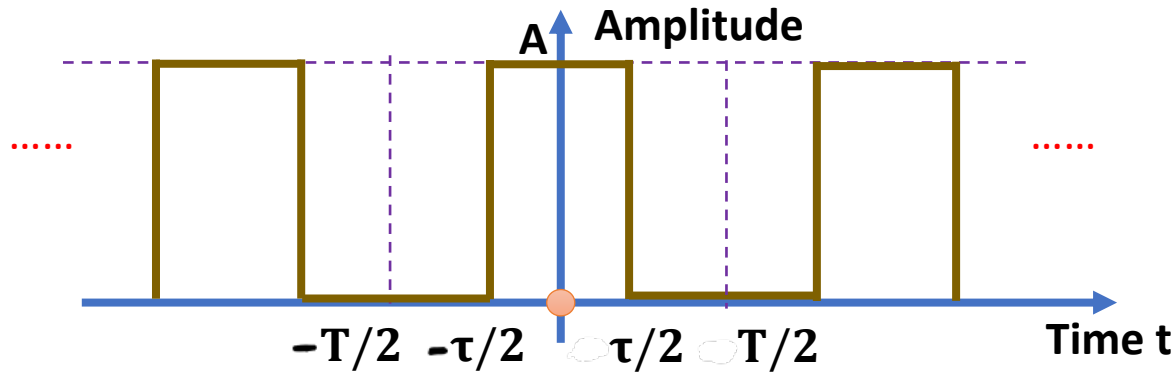
$$= a_0 + \sum_{n=1}^{\infty} a_n \cos(n(\omega_0 t + 2\pi)) + b_n \sin(n(\omega_0 t + 2\pi))$$

$$= a_0 + \sum_{n=1}^{\infty} a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t)$$

$$= x(t)$$

Example of Exponential Fourier Series

Example 1. Find C_n for signal $x(t)$:



Exponential Fourier Series:

$$\Rightarrow x(t) = \sum_{n=-\infty}^{\infty} C_n e^{jn\omega_0 t}$$

Exponential Fourier Series Coefficient C_n :

$$\begin{aligned}\Rightarrow C_n &= \frac{1}{T} \int_T x(t) e^{-jn\omega_0 t} dt \\ &= \frac{1}{T} \int_{-T/2}^{T/2} x(t) e^{-jn\omega_0 t} dt \\ &= \frac{1}{T} \int_{-\tau/2}^{\tau/2} A e^{-jn\omega_0 t} dt\end{aligned}$$

$$= \frac{A}{T} \left[\frac{e^{-jn\omega_0 t}}{-jn\omega_0} \right]_{-T/2}^{T/2}$$

$$= \frac{A}{T} \times \frac{1}{-jn\omega_0} \left[e^{-jn\omega_0 T/2} - e^{jn\omega_0 T/2} \right]$$

$$= \frac{2A}{Tn\omega_0} \left[\frac{e^{jn\omega_0 T/2} - e^{-jn\omega_0 T/2}}{j2} \right]$$

$$\left[\sin t = \frac{e^{jt} - e^{-jt}}{2j} \right]$$

$$\Rightarrow C_n = \frac{2A}{Tn\omega_0} \sin(n\omega_0 T/2)$$

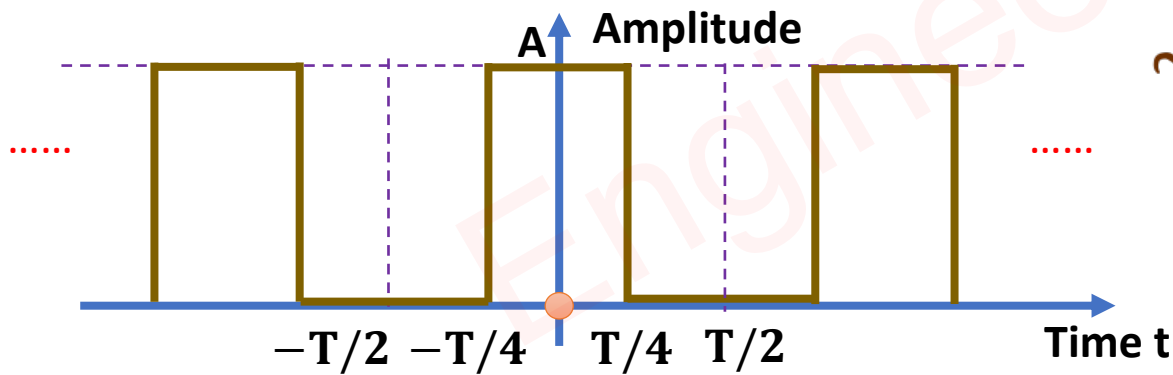
$$\left[\text{sinc}(t) = \frac{\sin t}{t} \right]$$

$$\Rightarrow C_n = \frac{2A}{T} \times \frac{T}{T} \times \frac{\sin(n\omega_0 T/2)}{n\omega_0 T/2} = \frac{AT}{T} \text{sinc}(n\omega_0 T/2)$$

Examples of Exponential Fourier Series

Example 1. Determine the Fourier Series Coefficient for the given periodic Signal $x(t)$:

~~a.~~ $\frac{A}{j\pi k} \sin\left(\frac{\pi}{2}k\right)$ $\Rightarrow C_{-k} = \frac{-A}{\pi k} \sin\left(\frac{\pi}{2}(-k)\right) = \frac{A}{\pi k} \sin\left(\frac{\pi}{2}k\right) = C_k$
~~b.~~ $\frac{A}{\pi k} \sin\left(\frac{\pi}{2}k\right)$ $\Rightarrow C_k = \frac{-A}{\pi k} \cos\left(\frac{\pi}{2}(-k)\right) = \frac{-A}{\pi k} \cos\left(\frac{\pi}{2}k\right) = -C_k$
~~c.~~ $\frac{A}{j\pi k} \cos\left(\frac{\pi}{2}k\right)$
d. $\frac{A}{\pi k} \cos\left(\frac{\pi}{2}k\right)$



$\Rightarrow x(t) = x(-t) \rightarrow$ Even & Real

$x(t) \rightarrow C_k$

- | | |
|----------------|----------------|
| - Even & Real | - Even & Real |
| - Even & Imag. | - Even & Imag. |
| - Odd & Real | - Odd & Imag. |
| - Odd & Imag | - Odd & Real. |

Example 2. The Signal $x(t)$ mapping with Exponential Fourier Series Coefficient is given as follows:

$$x(t) \leftrightarrow C_n = \begin{cases} 2, & n = 0 \\ j\left(\frac{1}{2}\right)^{|n|}, & \text{Otherwise} \end{cases}$$

$\Rightarrow C_n = C_{-n} \rightarrow \text{Even} \rightarrow x(t) \text{ also Even}$

$\downarrow$
 $\frac{dx(t)}{dt} \text{ (odd)}$

- ☒ a. $x(t)$ is a real valued Signal
- ☒ b. $x(t)$ is an even signal
- ☒ c. $dx(t)/dt$ is an even signal
- ☒ d. both option (b) and (c) are correct

Example 3. Suppose we have given following information about Signal $x(t)$:

- $x(t)$ is real Odd. $\longrightarrow$ Sine function
- $x(t)$ is periodic with duration 2 $\longrightarrow T = 2 \Rightarrow \omega_0 = \frac{2\pi}{T} = \frac{2\pi}{2} = \pi$ rad/sec
- Fourier Coefficients of $x(t) = 0$ for $|n| > 1$
 $\quad \quad \quad - \sin(\omega_0 t) = \sin(\pi t)$
- $\frac{1}{2} \int_0^2 |x(t)|^2 dt = 1$

The signal that satisfy these conditions is

$$\begin{array}{ccccccc} \cdots & c_{-2} & c_{-1} & c_0 & c_1 & c_2 & \cdots \\ \hline & = 0 & & \downarrow & & = 0 & \\ & & & = 0 & & & \end{array}$$

- ~~a.~~ $\sqrt{2} \sin(\pi t)$ & Unique
- ☒ b. $\sqrt{2} \sin(\pi t)$ but not Unique
- ~~c.~~ $2 \sin(\pi t)$ & Unique
- ~~d.~~ $2 \sin(\pi t)$ but not Unique

$$\Rightarrow P_{avg} = 1 = \frac{1}{2} \int_0^2 |x(t)|^2 dt$$

$$\Rightarrow 2 = \int_0^2 |x(t)|^2 dt$$

$$\Rightarrow A_m = \sqrt{2}$$